Assignment 11

`

```c
#include <stdio.h>
#include <math.h>

// Factorial function
long long factorial(int n) {
    long long fact = 1;
    for(int i = 1; i <= n; i++)
        fact *= i;
    return fact;
}

// Prime check function
int isPrime(int n) {
    if(n <= 1) return 0;
    for(int i = 2; i <= sqrt(n); i++) {
        if(n % i == 0)
            return 0;
    }
```

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```
}

// Prime factors function
void primeFactors(int n) {
    printf("Prime factors: ");
    for(int i = 2; i <= n; i++) {
        while(n % i == 0) {
            printf("%d ", i);
            n /= i;
        }
    }
    printf("\n");
}

int main() {
    int choice, num;

    printf("Enter a number: ");
    scanf("%d", &num);
```

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```
do {
    printf("\n--- MENU ---\n");
    printf("1. Square\n");
    printf("2. Square Root\n");
    printf("3. Cube\n");
    printf("4. Check Prime\n");
    printf("5. Factorial\n");
    printf("6. Prime Factors\n");
    printf("7. Exit\n");

    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Square = %d\n", num * num);
            break;
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    case 2:
        if(num >= 0)
            printf("Square Root = %.2f\n", sqrt(num));
        else
            printf("Invalid input for square root.\n");
        break;

    case 3:
        printf("Cube = %d\n", num * num * num);
        break;

    case 4:
        if(isPrime(num))
            printf("It is a Prime number\n");
        else
            printf("Not a Prime number\n");
        break;
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```
    case 5:
        if(num < 0)
            printf("Factorial not defined\n");
        else
            printf("Factorial = %lld\n", factorial(num
                ));
        break;

    case 6:
        if(num > 1)
            primeFactors(num);
        else
            printf("No prime factors\n");
        break;

    case 7:
        printf("Exiting...\n");
        break;
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        case 7:
            printf("Exiting...\n");
            break;

        default:
            printf("Invalid choice\n");
    }

    } while(choice != 7);

    return 0;
}
```

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output-

```
Enter a number: 2

--- MENU ---
1. Square
2. Square Root
3. Cube
4. Check Prime
5. Factorial
6. Prime Factors
7. Exit
Enter choice: 1
Square = 4

--- MENU ---
1. Square
2. Square Root
3. Cube
4. Check Prime
5. Factorial
```

```
6. Prime Factors
7. Exit
Enter choice: 2
Square Root = 1.41

--- MENU ---
1. Square
2. Square Root
3. Cube
4. Check Prime
5. Factorial
6. Prime Factors
7. Exit
Enter choice: 3
Cube = 8

--- MENU ---
1. Square
2. Square Root
```

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```
Square Root
Cube
Check Prime
Factorial
Prime Factors
Exit
:er choice: 4
is a Prime number

- MENU ---
Square
Square Root
Cube
Check Prime
Factorial
Prime Factors
Exit
:er choice: 5
:torial = 2
```

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`

```
1. Square
2. Square Root
3. Cube
4. Check Prime
5. Factorial
6. Prime Factors
7. Exit
Enter choice: 6
Prime factors: 2

--- MENU ---
1. Square
2. Square Root
3. Cube
4. Check Prime
5. Factorial
6. Prime Factors
7. Exit
Enter choice: 7
```